

■ ELECTRONIC CONTROL SYSTEM

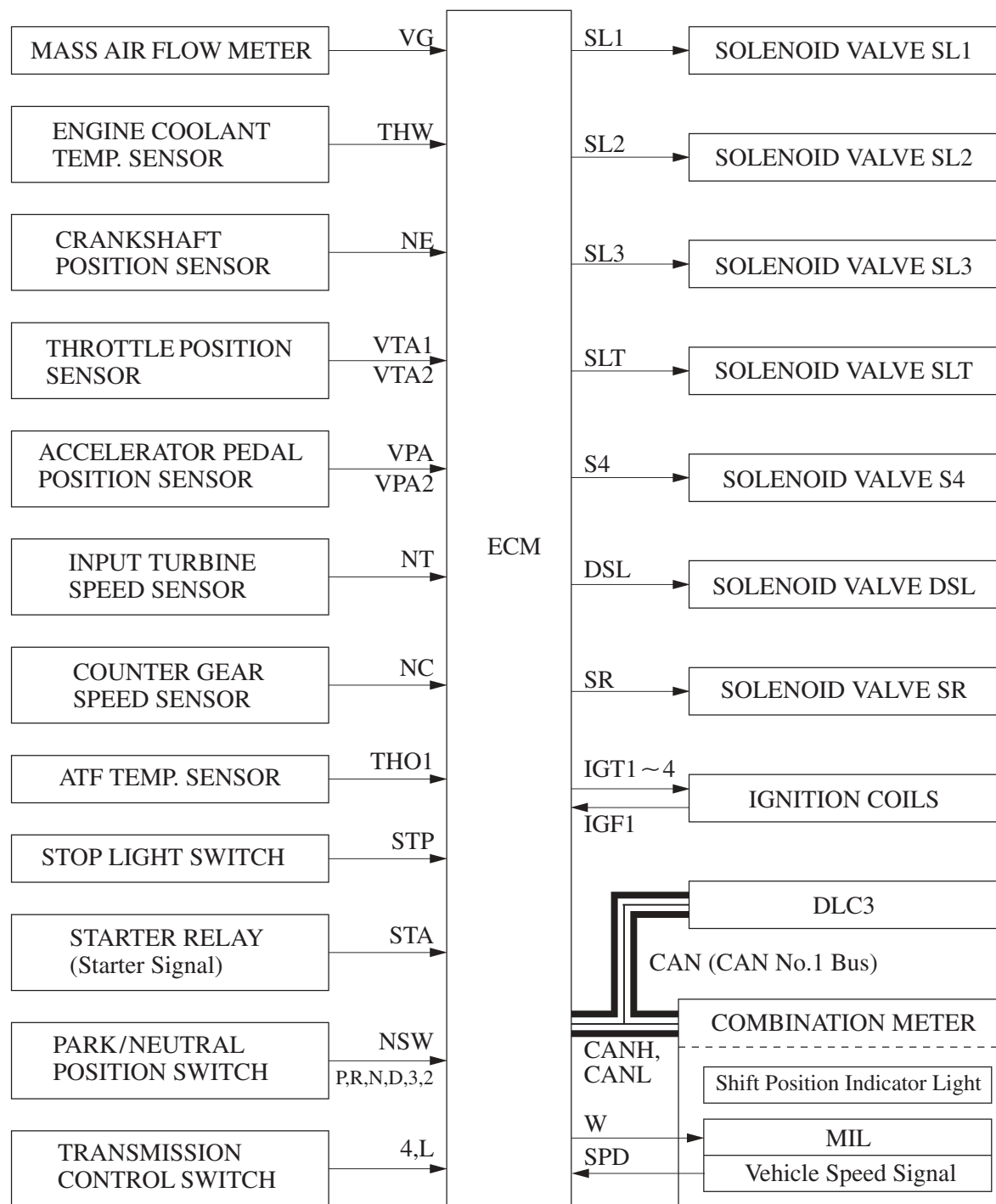
1. General

The electronic control system of the U250E automatic transaxle consists of the control listed below.

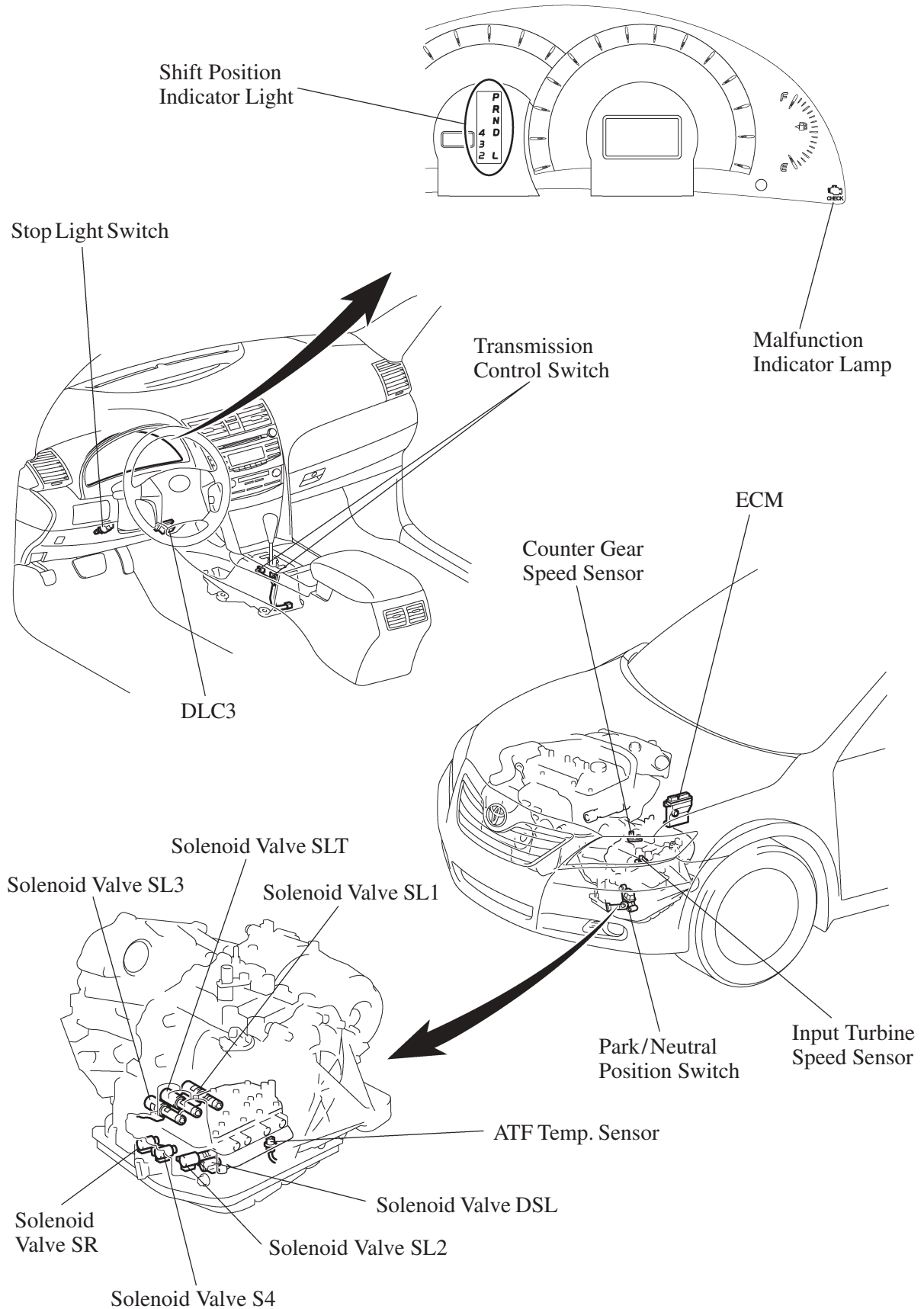
System	Outline
Shift Timing Control	The ECM sends current to 3 solenoid valves (SL1, SL2, and SL3) based on signals from each sensor and shifts the gear.
Clutch Pressure Control (See page CH-28)	- Controls the pressure that is applied directly to B₁ brake, C₀ and C₁ clutches by actuating 3 solenoid valves (SL1, SL2, and SL3) in accordance with ECM signals. 3 solenoid valves (SL1, SL2, and SL3) minutely control the clutch pressure in accordance with the engine output and driving conditions.
Line Pressure Optimal Control (See page CH-29)	Actuates the solenoid valve SLT to control the line pressure in accordance with information from the ECM and the operating conditions of the transaxle.
Shifting Control in Uphill/ Downhill Traveling (See page CH-30)	Controls to restrict the 4th or 5th upshift or to provide appropriate engine braking by the ECM to determine whether the vehicle is traveling uphill or downhill.
Lock-up Timing Control	The ECM sends current to the solenoid valves DSL and SL2 based on signals from each sensor and engages or disengages the lock-up clutch.
Flex Lock-up Clutch Control (See page CH-31)	Controls the solenoid valve SL2 and DSL, provides an intermediate mode between the ON/OFF operation of the lock-up clutch, and increase the operating range of the lock-up clutch to improve fuel economy.
Engine Torque Control	Retards the engine ignition timing temporarily to improve shift feeling during up or down shifting.
“N” to “D” Squat Control	When the shift lever is shifted from “N” to “D” position, the gear is temporarily shifted to 3rd and then to 1st to reduce vehicle squat.
Diagnosis (See page CH-32)	When the ECM detects a malfunction, the ECM makes a diagnosis and memorizes the malfunctioning part.
Fail-safe (See page CH-32)	Even if a malfunction is detected in the sensors or solenoids, the ECM activates fail-safe control to prevent the vehicle’s drivability from being significantly affected.

2. Construction

The configuration of the electronic control system in the U250E automatic transaxle is as shown in the following chart.



3. Layout of Main Components

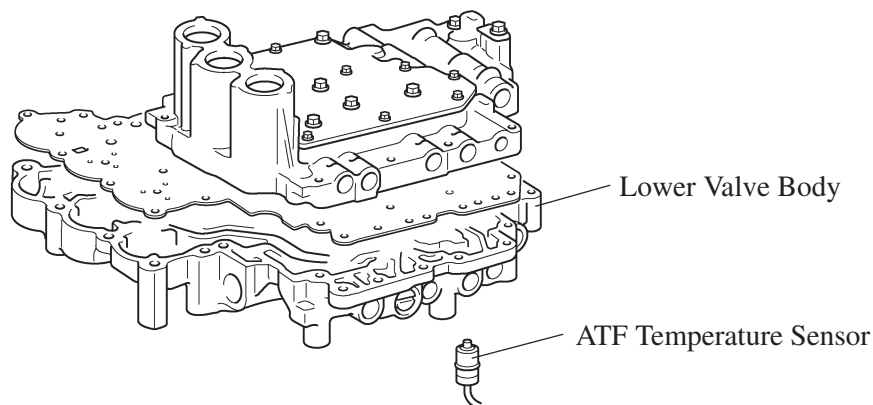


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4. Construction and Operation of Main Components

ATF Temperature Sensor

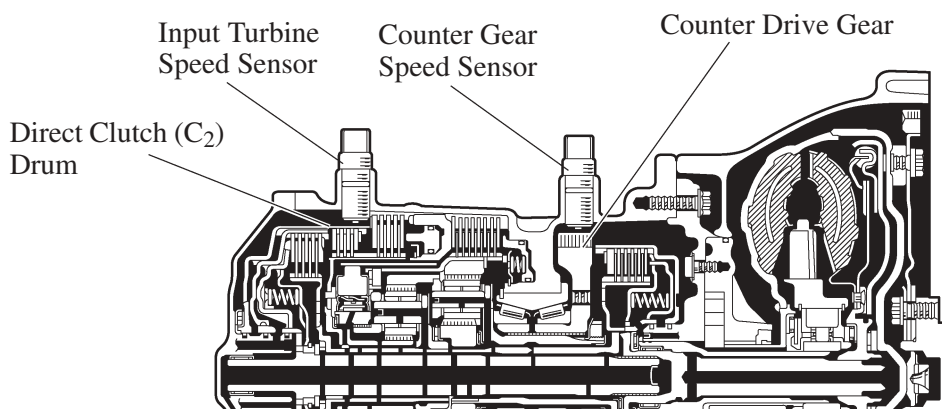
- The ATF temperature sensor is installed in the valve body for direct detection of the fluid temperature.
- The ATF temperature sensor is used for the revision of clutch and brake pressures to maintain a smooth shift quality every time.



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Speed Sensors

- The U250E automatic transaxle uses an input turbine speed sensor (for the NT signal) and a counter gear speed sensor (for the NC signal). Thus, the ECM can detect the timing of the shifting of the gears and appropriately control the engine torque and hydraulic pressure in response to the various conditions. These speed sensors are the pick-up coil type.
- The input turbine speed sensor detects the input speed of the transaxle. The direct clutch (C_2) drum is used as the timing rotor for this sensor.
- The counter gear speed sensor detects the speed of the counter gear. The counter drive gear is used as the timing rotor for this sensor.



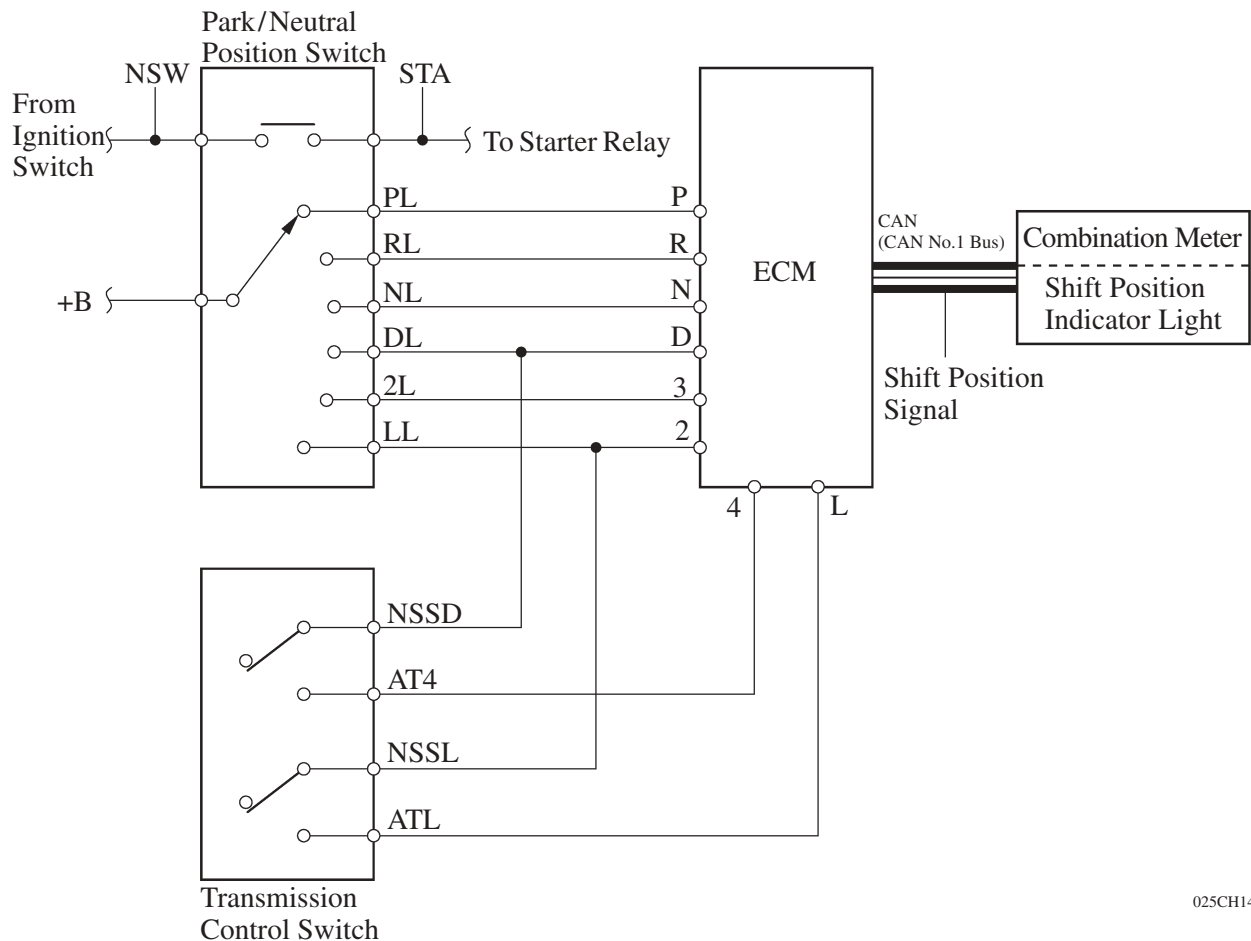
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Transmission Control Switch and Park/Neutral Position Switch

The ECM uses these switches to detect the shift lever position.

- The park/neutral position switch sends the P, R, N, D, 3 and 2 position signals to the ECM.
- The transmission control switch is installed inside the shift lever assembly. This switch sends the 4 and L position signals to the ECM.
- The ECM sends these shift position signals to the combination meter (meter ECU) through CAN communication. The combination meter (meter ECU) controls the shift position indicator light based on these signals.

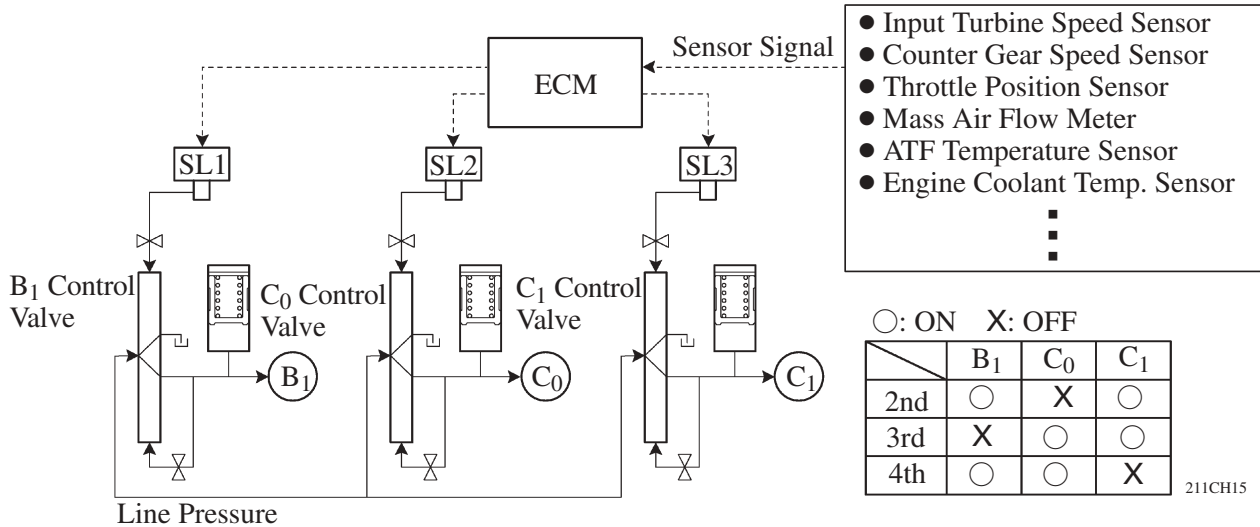
► Wiring Diagram ◀



5. Clutch Pressure Control

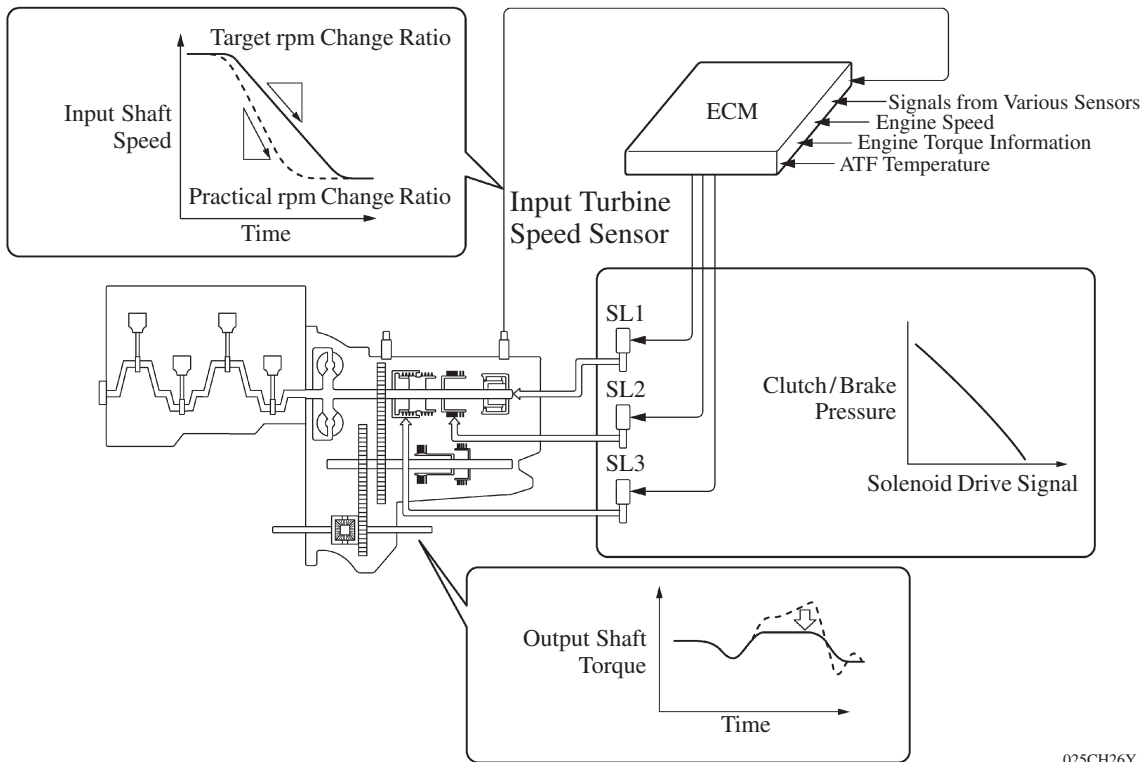
Clutch to Clutch Pressure Control

- This control is used for shifting from the 2nd to 3rd gear, and from the 3rd to 4th gear.
- Solenoid valves actuate SL1, SL2, and SL3 in accordance with the signals from the ECM, and guide this output pressure directly to the control valves B₁, C₀, and C₁ in order to regulate the line pressure that acts on the B₁ brake, C₀ and C₁ clutches. As a result, compact B₁, C₀ and C₁ accumulators without a back pressure chamber have been realized.



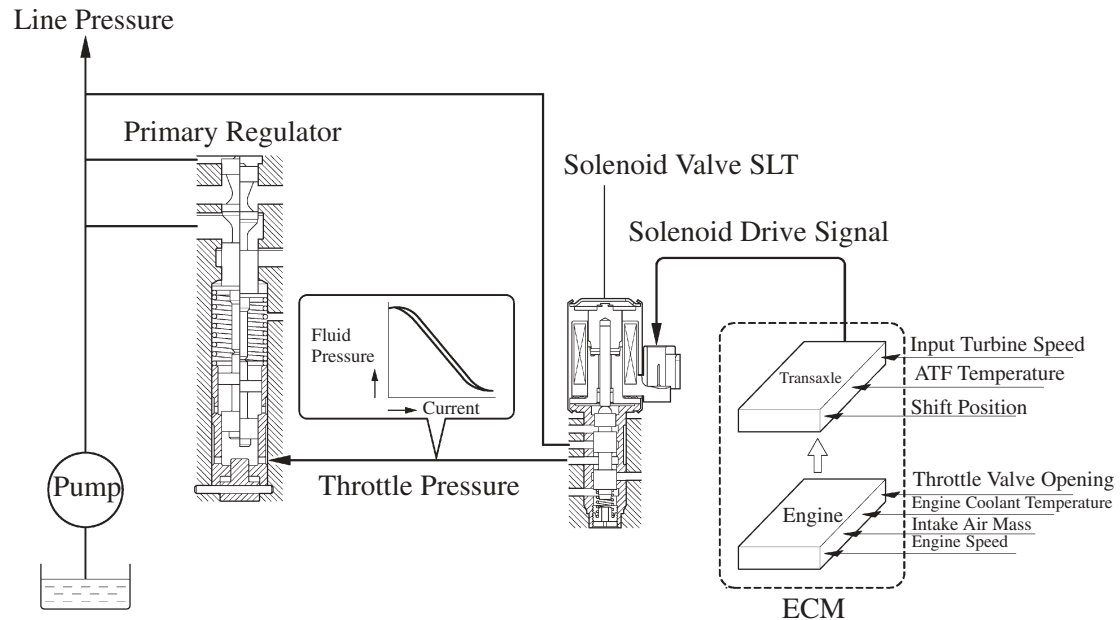
Clutch Pressure Optimal Control

The ECM monitors the signals from various types of sensor such as the input turbine speed sensor, allowing shift solenoid valves SL1, SL2, and SL3 to minutely control the clutch pressure in accordance with engine output and driving conditions. As a result, smooth shift characteristics have been realized.



6. Line Pressure Optimal Control

The line pressure is controlled by using solenoid valve SLT. Through the use of solenoid valve SLT, the line pressure is optimally controlled in accordance with the engine torque information, as well as with the internal operating conditions of the torque converter and the transaxle. Accordingly, the line pressure can be accurately controlled in accordance with the engine output, traveling condition, and the ATF temperature, thus realizing smooth shift characteristics and optimizing the workload of the oil pump.



7. Shifting Control in Uphill/Downhill Traveling

General

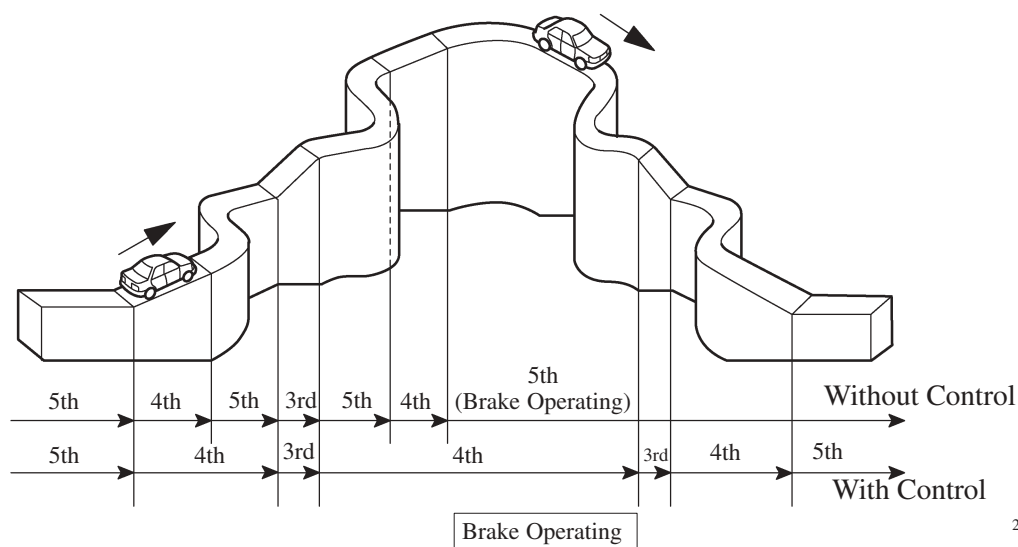
This control helps minimize the shifting of gears when the driver operates the accelerator pedal while driving on a winding road with ups and downs, in order to ensure a smooth drive.

Shift Control in Uphill Traveling

- When the ECM determines uphill travel, it prohibits the transaxle from shifting up into 5th after the transaxle has shifted down below 4th.
- When the ECM determines uphill travel with a steeper grade, it prohibits the transaxle from shifting up into 4th after the transaxle has shifted down below 3rd.

Shift Control in Downhill Traveling

- When the ECM determines downhill travel, it shifts down the transaxle from 5th to 4th in accordance with the brake operation signal that is input when the driver operates the brake pedal.
- When the ECM determines downhill travel with a steeper grade, and a brake operation signal is input again, the ECM shifts the transaxle down from 4th to 3rd.

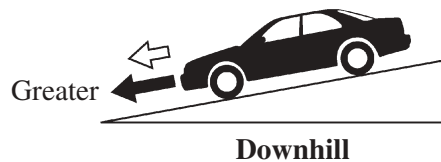
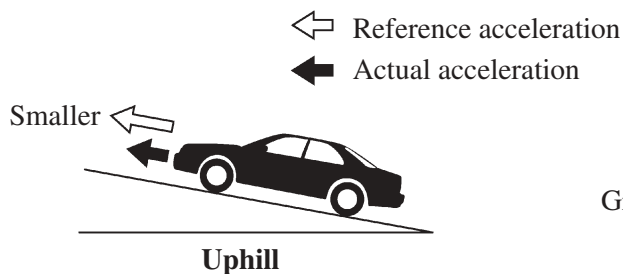


Uphill/Downhill Judgment

The actual acceleration calculated from the speed sensor signal is compared with the reference acceleration stored in the ECM to judge uphill or downhill traveling.

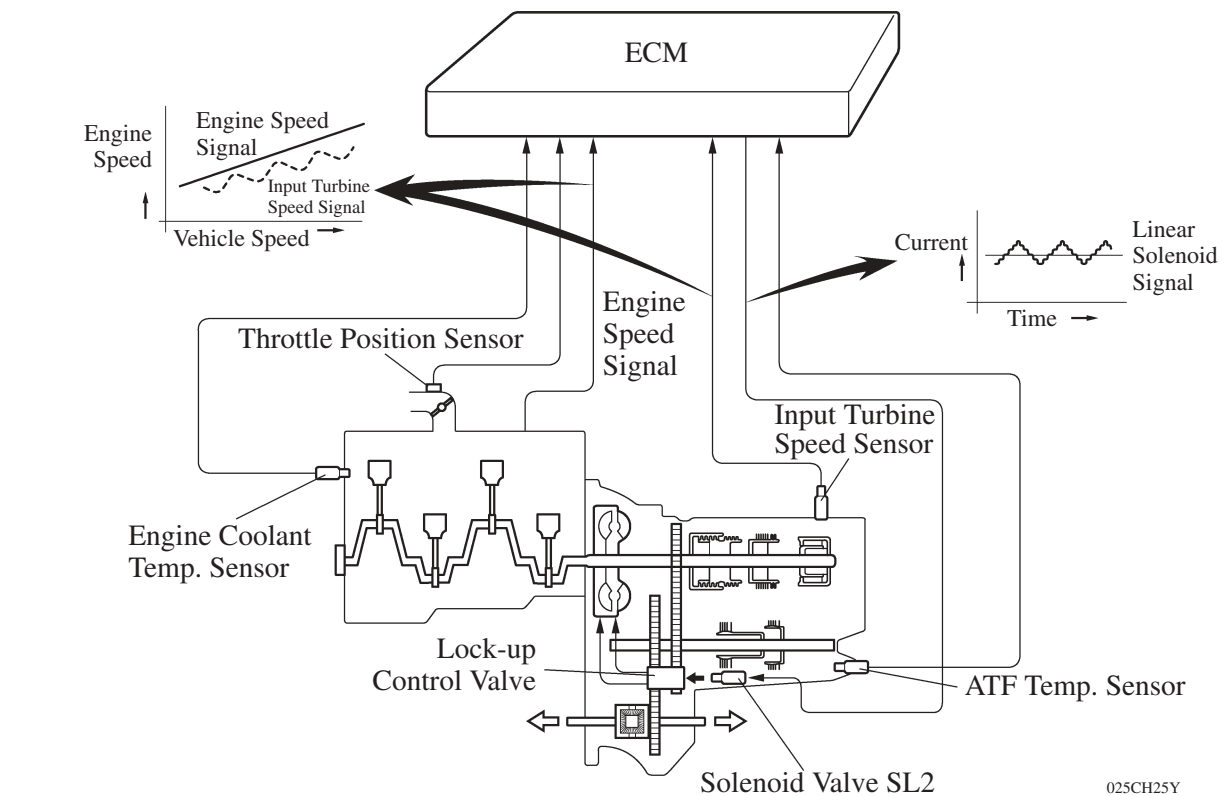
Actual Acceleration < Reference Acceleration

Actual Acceleration > Reference Acceleration



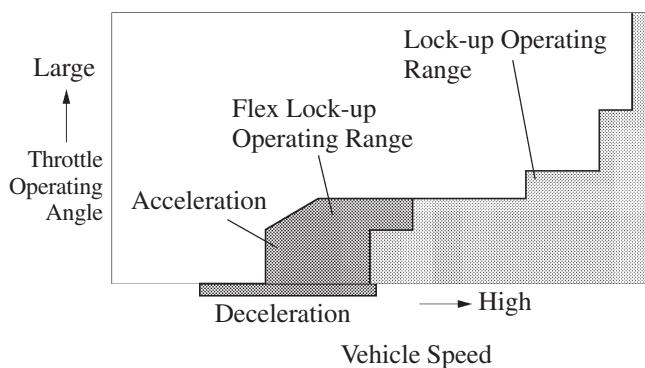
8. Flex Lock-up Clutch Control

- In addition to the conventional lock-up timing control, flex lock-up clutch control is used.
- This flex lock-up clutch control regulates solenoid valve SL2 as an intermediate mode between the ON and OFF operations of the lock-up clutch.
- The flex lock-up clutch control operates during acceleration, in the 4th and 5th gear in the D range, and during deceleration, in the 4th and 5th gear in the D range, and in the 4th gear in the 4 range.
- During acceleration, the partition control of the power transmission between the lock-up clutch and torque converter greatly boosts the transmission efficiency in accordance with the driving conditions, improving the fuel economy.
- During deceleration, the lock-up clutch is made to operate. Therefore the fuel-cut area is expanded and fuel economy is improved.



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► Flex Lock-up Operation ◀



Flex Lock-up Timing

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Range	Gear	Acceleration Flex Lock-up	Deceleration Flex Lock-up
D	1st	X	X
	2nd	X	X
	3rd	X	X
	4th	○	○
	5th	○	○
4	1st	X	X
	2nd	X	X
	3rd	X	X
	4th	X	○

○: Operates X: Does not operate

9. Diagnosis

- When the ECM detects a malfunction, the ECM makes a diagnosis and memorizes the information related to the fault. Furthermore, the MIL (Malfunction Indicator Lamp) in the combination meter illuminates or blinks to inform the driver of the malfunction.
- At the same time, the DTC (Diagnosis Trouble Code) are stored in the memory. The DTC can be read by connecting a hand-held tester.
- For details, see the 2007 Camry Repair Manual (Pub. No. RM0250U).

Service Tip

The ECM uses the CAN protocol for diagnostic communication. Therefore, a hand-held tester and a dedicated adapter [CAN VIM (Vehicle Interface Module)] are required for accessing diagnostic data. For details, see the 2007 Camry Repair Manual (Pub. No. RM0250U).

10. Fail-safe

This function minimizes the loss of operation when any abnormality occurs in a sensor or solenoid.

► Fail-safe Control List ◀

Malfunction Part	Function
Speed Sensor	During a speed sensor malfunction, the vehicle speed is detected through the signals from the counter gear speed sensor to effect normal control.
Counter Gear Speed Sensor	During a counter gear speed sensor malfunction, 5th upshift is prohibited.
ATF Temp. Sensor	During a ATF temperature sensor malfunction, 5th upshift is prohibited.
Solenoid Valve SL1, SL2, SL3, and S4	The current to the failed solenoid valve is cut off and control is effected by operating other solenoid valves with normal operation. Shift control is effected as described in the table in the next page, depending on the failed solenoid. Even if the engine starts under this condition, the gear position remains where it was before.

○: ON Δ: According to Flex Lock-up X: OFF

Normal	Solenoid Valve	SL1	○	X	○	X	X
		SL2	○	○	X	Δ	Δ
		SL3	X	X	X	○	○
		S4	X	X	X	X	○
	Gear Position		1st	2nd	3rd	4th	5th
SL1 Malfunction (During driving at 1st or 2nd)	Solenoid Valve	SL1	X				
		SL2	○	○	○	○	○
		SL3	X	X	X	○ → X	○ → X
		S4	X	X	X	X	○ → X
	Gear Position		1st → 2nd	2nd	3rd → 2nd	4th → 2nd	5th → 2nd
SL1 Malfunction (During driving at 3rd)	Solenoid Valve	SL1	X				
		SL2	○ → Δ	○ → Δ	Δ	Δ	Δ
		SL3	X	X	X	○ → X	○ → X
		S4	X → ○	X → ○	X → ○	X → ○	○
	Gear Position		1st → 4th	2nd → 4th	3rd → 4th	4th	5th → 4th
SL1 Malfunction (During driving at 4th or 5th)	Solenoid Valve	SL1	X				
		SL2	○ → Δ	○ → Δ	Δ	Δ	Δ
		SL3	X → ○	X → ○	X → ○	○	○
		S4	X	X	X	X	○
	Gear Position		1st → 4th	2nd → 4th	3rd → 4th	4th	5th
SL2 Malfunction	Solenoid Valve	SL1	○	X → ○	○	X → ○	X → ○
		SL2	X				
		SL3	X	X	X	○ → X	○ → X
		S4	X → ○	X → ○	X → ○	X → ○	○
	Gear Position		1st → 4th	2nd → 4th	3rd → 4th	4th	5th → 4th
SL3 Malfunction	Solenoid Valve	SL1	○	X	○	X → ○	X → ○
		SL2	○	○	X	Δ	Δ
		SL3	X				
		S4	X	X	X	X → ○	○
	Gear Position		1st	2nd	3rd	4th	5th → 4th
S4 Malfunction	Solenoid Valve	SL1	○	X	○	X	X
		SL2	○	○	X	Δ	Δ
		SL3	X	X	X	○	○
		S4	X				
	Gear Position		1st	2nd	3rd	4th	5th → 4th
SL1, SL2, SL3, and S4 Malfunction	Solenoid Valve	SL1	X				
		SL2	X				
		SL3	X				
		S4	X				
	Gear Position		1st → 4th	2nd → 4th	3rd → 4th	4th	5th → 4th